



COURSE NUMBER: CR1511

COURSE TITLE: Website Development

COURSE DESCRIPTION:

This course will introduce students to the key concepts of HTML and CSS. The student will begin creating basic web pages and move on to creating full websites. The student will then enhance web pages with more advanced layouts, embedded fonts, audio, video and responsive design. The student will use basic web design principles and create a website using the content management system WordPress.

PREREQUISITES: None

CO-REQUISITES: None

CREDIT VALUE: Three (3)

COURSE HOURS PER WEEK: Two (2)

LAB HOURS PER WEEK: Two (2)

REQUIRED TEXT:

Ruvalcabe, A.B. (2018). *Murach's HTML5 and CSS3* (4th Edition). Mike Murach & Associates, Inc. eText ISBN: 9781943872305

Ruvalcaba Z, Boehm A, Delamater M. (2024) *Murach's HTML and CSS (6th Edition)*. Mike Murach & Associates, eText ISBN: 9781943873227, Print ISBN 9781943873210

LEARNING RESOURCES:

W3C Markup Validation Service <https://validator.w3.org>

MAJOR TOPICS:

- 1.0 Introduction to HTML5
- 2.0 Cascading Style Sheets
- 3.0 Enhancing a Web Page
- 4.0 Website Design
- 5.0 WordPress Websites

LEARNING OBJECTIVES:

1.0 Introduction to HTML5

- 1.1 Web Development
 - 1.1.1 Load a webpage from the internet or local server into a web browser
 - 1.1.2 View the source code for a web page in a web browser
 - 1.1.3 Use a browsers development tools to examine a web page
 - 1.1.4 Use a text editor or IDE to create and edit HTML and CSS files
 - 1.1.5 Test HTML documents by loading them into your browser
 - 1.1.6 Validate a HTML document by using a Markup validator. (Like the W3C Markup Validation Service)
- 1.2 Create Web Pages
 - 1.2.1 Create a properly structured HTML page using a mix of semantic and non-semantic elements
 - 1.2.2 Build the relative URLs for any file in the folder structure of a website

2.0 Cascading Style Sheets

- 2.1 Create a CSS style sheet for formatting a web page using selectors
- 2.2 Use developer tools in the browser to inspect the styles of a webpage
- 2.3 Determine the HTML being referenced by a CSS selector
- 2.4 Create a CSS style rule that uses the box model
 - 2.4.1 Apply the following to a webpage:
 - 2.4.1.1 spacing
 - 2.4.1.2 borders
 - 2.4.1.3 backgrounds
- 2.5 Create a stylesheet that uses floating to implement a fixed page layout with multicolumn design with headers and footers
- 2.6 Create a stylesheet that uses floating to implement a fluid page layout with multicolumn design with headers and footers
- 2.7 Format links with CSS
- 2.8 Format lists with CSS
- 2.9 Convert a webpage that uses a fixed layout so that it uses a fluid layout
- 2.10 Modify a webpage that uses a fluid layout to make it responsive based on the size of the screen where it is displayed
- 2.11 Update media queries so that a different menu is used when a webpage is displayed on a smaller screen
- 2.12 Use browser developer tools to test a web pages responsive design

3.0 Enhancing a Web Page

- 3.1 Apply Flexbox and Grid layouts
 - 3.1.1 Use Flexbox layout to lay out webpages

- 3.1.2 Use Flexbox layout to layout components within webpages
 - 3.1.3 Use Grid layout to lay out webpages
 - 3.1.4 Use Grid layout to lay out components within webpages
 - 3.1.5 Use Flexbox layouts and Grid layouts on mobile devices
- 3.2 Work with images and icons
 - 3.2.1 Use HTML to include images on a web page
 - 3.2.2 Use CSS to format the images on a web page
 - 3.2.3 Create an image map
 - 3.2.4 Use the image map so that it links to more than one webpage
- 3.3 Work with tables and forms
 - 3.3.1 Create HTML tables with merged cells
 - 3.3.2 Create HTML tables with caption headers
 - 3.3.3 Create HTML tables with caption footers
 - 3.3.4 Format the HTML table with CSS structural pseudo-classes
 - 3.3.5 Use techniques to enhance accessibility for tables
 - 3.3.6 Make the tables responsive
 - 3.3.7 Create a HTML form that includes various form controls
 - 3.3.8 Use CSS to format forms
 - 3.3.9 Use CSS to format form controls
 - 3.3.10 Use the html attributes with CSS selectors for data validation
- 3.4 Adding audio and video
 - 3.4.1 Add audio to a web page
 - 3.4.2 Add video to a web page
- 3.5 Use fonts, transitions, transforms, animations and filters
 - 3.5.1 Use the @font-face selector to embed fonts in a webpage
 - 3.5.2 Use the @media print selector to add the styles for printing a page to the style sheet for displaying the page
 - 3.5.3 Use a second style sheet that is only used when a page is being printed

4.0 Website Design

- 4.1 Design a website that is responsive with usability in mind
- 4.2 Implement a design that helps users know where they are on a website
- 4.3 Apply the three guidelines for typography on a webpage
- 4.4 Create a site map when designing a webpage
- 4.5 Use a wireframe prototype for designing a webpage
- 4.6 Describe the lifecycle of a website
- 4.7 Describe why it is important to make a website easy to maintain

5.0 WordPress Websites

- 5.1 Describe WordPress

- 5.2 Detail the WordPress system requirements
- 5.3 Install webserver software such as laragon, wampserver or others
- 5.4 Install WordPress
- 5.5 Install a new WordPress theme to a WordPress site
- 5.6 Make a WordPress theme the default theme for a WordPress site
- 5.7 Customize a WordPress theme using the tools provided by the theme
- 5.8 Install a plugin to a WordPress site
- 5.9 Activate a plugin to a WordPress site
- 5.10 Use a plugin within a WordPress site
- 5.11 Add content to a WordPress site
- 5.12 Create the navigation menu for your WordPress site

EVALUATION:

Assignments	30%
Test(s)	20%
Project	20%
Final Exam	30%

DATE DEVELOPED: September 2020

DATE REVIEWED:

REVISION NUMBER:

DATE REVISED:

Effective PeopleSoft Date: September 2025

Note to instructor: Check PIRS to ensure this outline is the most current version.